

Shaotian Sun

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EDUCATION

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| University of Michigan, Ann Arbor
<i>B.S. in Honors Mathematics, Minor in CS</i> | Aug. 2024 – Dec. 2026
<i>Ann Arbor, MI</i> |
| ◦ Cumulative GPA: 3.97/4.00 | |
| Shanghai Jiao Tong University
<i>B.Eng. in Electrical and Computer Engineering</i> | Sept. 2022 – Aug. 2027
<i>Shanghai, China</i> |
| ◦ Cumulative GPA: 3.75/4.00 (Top 10%) | |

PREPRINT

(* indicates equal contribution.)

Yifan Jing*, Emma Yicheng Lou*, **Shaotian Sun***, Yuxuan Wu*. “Computing Tools for Translation-Invariant Total Orders.” [arXiv link]

RESEARCH EXPERIENCES

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| Conditional Exponential Directed Last Passage Percolation with Flat Initial Condition (2026 Umich REU)
<i>Supervised by Prof. Jinho Baik</i> | May 2026 - Present
<i>Dept. of Math, University of Michigan</i> |
| ◦ Investigated conditional exponential DLPP (directed last passage percolation) with flat initial condition under a one-point upper large-deviation event. | |
| ◦ Conjectured the law of large number for conditional exponential DLPP heuristics. | |
| ◦ Proved conditional near-diagonal fluctuation theorem using method of steepest descent, and heuristically extended to fluctuations for off-diagonal points. | |
| High-Dimensional Latent Pairwise Comparison Model
<i>Supervised by Prof. Gongjun Xu</i> | Apr. 2026 - Present
<i>Dept. of Stats, University of Michigan</i> |
| ◦ Investigate statistical estimation and identifiability in the Blade-Chest-Inner Model with low-rank latent interaction structures. | |
| ◦ Working on both average and uniform consistency and asymptotic distribution results. | |
| Computing Tools for Translation-Invariant Total Orders (2025 Umich Lab of Geometry)
<i>Supervised by Dr. Grant Barkley</i> | Aug. 2025 - Jul. 2026
<i>Dept. of Math, University of Michigan</i> |
| ◦ Designed finite representations and graph-based algorithms for comparing TITOs (Translation-Invariant Total Orders), generating local Hasse Diagrams in small n cases, and computing the join of TITOs. | |
| ◦ Implemented them in the Python package TITO_Explore. [Github] | |
| Provable In-Context Learning and Gradient Descent in Recurrent Neural Networks (2024 SJTU PRP)
<i>Supervised by Prof. Shuai Li</i> | Feb. 2024 - Nov. 2024
<i>Shanghai Jiao Tong University</i> |
| ◦ Developed a constructive theoretical framework showing how Transformer attention and MLP layers can implement gradient-descent updates for a two-layer RNN (Recurrent Neural Network) by sequentially representing its hidden-state and output computations. | |

TALKS & PRESENTATIONS

- Conditional Exponential DLPP with Flat Initial Condition** [Slides][Seminar Recording (38:29 - 1:04:17)]
- Young Mathematician Conference, Columbus, OH, 2026.
 - University of Michigan REU Seminar, Ann Arbor, MI, 2026.
- TITography: When Order Takes Shape** [Poster]
- Undergraduate Research Symposium, Ann Arbor, MI, 2026.

- Lab of Geometry Final Poster Session, Ann Arbor, MI, 2025.

Algorithmic Construction of Weak Orders between Translation-Invariant Total Orders (TITOs) [Slides]

- Lab of Geometry Midterm Presentation, Ann Arbor, MI, 2025.

Is Factorization Always Unique? A Journey into Algebraic Number Theory [Slides]

- Directed Reading Program Presentation Session, Ann Arbor, MI, 2025.

A Prototype of Martian Terrain Detecting System [Poster] [Slides]

- 2023 Summer Design Expo, Shanghai, China, 2023.
- VG100 Symposium, Shanghai, China, 2023.

CONFERENCES, WORKSHOPS & SEMINARS ATTENDED

2026 University of Michigan REU Seminar, University of Michigan, Ann Arbor, MI

2026 University of Michigan Random Matrix Theory Summer School, University of Michigan, Ann Arbor, MI
UP GRADe, University of Pennsylvania, Philadelphia, PA

2026 Byrne Conference on Stochastic Analysis in Finance and Insurance, University of Michigan, Ann Arbor, MI

SKILLS

Programming: Python (Pytorch), L^AT_EX, C++, STATA, Matlab, C, Git, Linux

Languages: Chinese (native), English (TOEFL: 110 [R30/L27/S25/W28]), German (beginner)

RELEVANT COURSEWORK

Graduate-Level Math: Probability Theory (A+), Measure Theory and Functional Analysis (A), Complex Analysis (A), Graph Theory (A+), Commutative Algebra (A).

Undergraduate Math: Linear Algebra (A), Honors Multivariate Analysis (A), Honors Differentiable Manifolds (A+), Honors Abstract Algebra II (A+), Honors Abstract Algebra I (A), Single Variable Analysis (A+), Differential Equations (A), Discrete Math (A+).

Stats & Computing: Probability & Statistics in Engineering (A+), Econometrics (A), Machine Learning (A), Web Database (A-), Data Analysis (A+), Theoretical Computer Science (A), Data Structure and Algorithm (A), Signal and System (A).

DIRECTED & INDEPENDENT READING

Representation Theory: A First Course (Harris & Fulton)

Mentor: Prof. Griffin Wang

Aug. 2025 - Apr. 2026

Independent Reading

Number Fields (Marcus)

Mentor: Ritwick Bhargava

May 2025 - Aug. 2025

Directed Reading Program

TEACHING EXPERIENCES

Course Assistant, MATH 201: Introduction to Mathematical Writing, Winter 2026, *University of Michigan*

Kiluk Tutor, MATH 217: Linear Algebra, Winter 2026, *University of Michigan*

Grader, MATH 525: Probability Theory, Fall 2025, *University of Michigan*

Grader, MATH 217: Linear Algebra, Winter 2025, *University of Michigan*

Teaching Assistant, TC 3000J: Technical Communication, Summer 2024, *Shanghai Jiao Tong University*

Teaching Assistant, TC 3000J: Technical Communication, Spring 2024, *Shanghai Jiao Tong University*

Teaching Assistant, ENGL 1000J: Academic Writing I, Fall 2023, *Shanghai Jiao Tong University*

LEADERSHIP & OUTREACH

Student Advisor, SJTU Global College Advising Center, 2024–2026, *Shanghai Jiao Tong University*

Community Assistant, U(M) Math Corps, 2025, *University of Michigan*

Interview Group Leader, SJTU Global College Interview Group, 2022–2024, *Shanghai Jiao Tong University*

HONORS & AWARDS

Jack McLaughlin Award in Algebra	2026
James B. Angell Scholar	2026
M.S. Keeler Dept. of Math Merit Scholarship	2025 - 2026
University Honors	2024 - 2026
National Undergraduate Scholarship, China	2023 - 2024
Best Innovation Award in Summer Design Expo	2023